QP CODE: 25901630

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course(DSC)–MG2DSCCND100- FUNDAMENTALS OF FOOD SCIENCE

(2024 ADMISSION ONWARDS)

Duration:2 Hours **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S),Interest (I) and Appreciation(Ap)

Students should attempt atleast one question from each course outcome to enhance their overall outcome attainability.

Part A
Multiple Choice Questions
Answer all questions
Each question carries **1** mark

1. c) Roasting of coffee beans
2. a) Lipase
3. d) Foam
4. b). Thiamine
5. c) Poor portion control
6. d). None of these

Part B
Short Answer Type Questions.
Answer any **2** questions.
Each question carries **2** marks.

7. Summarise the role of serving size and recipe yield in standardization.

- **Serving Size (1 mark)**

Defining the exact portion size ensures consistency in meal service, proper nutrition distribution, and accurate cost control.
A single serving should be easy to replicate across multiple servings.
For example, "One serving = 200 g dal with 150 g rice."

- **Recipe Yield (1 mark)**

Recipe yield refers to the total quantity produced, either by weight or number of servings. Understanding yield helps in scaling the recipe up or down for different production requirements.
This information is essential for inventory planning and cost calculation.
For instance, a standard Vegetable Pulao recipe might yield 10 servings, each of 250 g.

8. Explain any four advantages of cutting. [K] / [CO2] .**0.5 Mark for each point (Any 4)**

- The product is easily cooked
- Easily consumed
- Spoiled portion can be discarded
- Appearance can be improved
- Helps in proper blending of the products.

9. Describe the scope of food science in nutrient profiling. [K] / [CO1] (**1 Mark for each point)**

- Food science helps to analyze and understand the nutrient content of foods, identifying the amounts of essential nutrients, vitamins, and minerals.
- It ensures that food products are nutritionally balanced and contribute positively to health when consumed.

10.List any 2 advantages and disadvantages of soaking. [K] / [CO2]

Advantages: 0.5 mark for each point (Any 2)

- Hastens the cooking process
- Facilitates grinding
- Softens food texture
- Peel can be easily removed from soaked products
- Make the pulp extraction easier
- reduces pungency of onion
- It initiates fermentation process

Disadvantages: 0.5 mark for each point

- water soluble nutrients may be leached into the water
- It involves time

Part C

Short Essay Type Questions.
Answer any**4** questions.
Each question carries **5** marks.

11. Explain the importance of food science in the modern society. [U] / [CO1]

- Ensuring Food Safety and Food Security.
- Innovation in the World of Food
- Offers a Wide Range of Jobs in the Field
- Prevent Money Loss of Food Processing Companies
- Transform Raw Materials into Finished, Delicious Product
- Food Science Makes Life Easy
- Regulating food Supply
- Waste Utilization and Reduction in Environment Pollution (**Any 5 points,1 mark for each point)**

12. Classify fermentation and explain its advantages and disadvantages. [U] / [CO2]

Classification (1 Mark)

- Alcoholic
- Lactic acid
- Acetic acid
- Alkaline **(0.25 Mark for each point.)**

Advantages (2 marks. Any 4 points)

- Texture and taste improved
- Protein quality improved
- B vitamins are synthesized by micro-organisms.
- Shelf life can be increased
- Easily digestible **(0.5 Mark for each point.)**

Disadvantages (2 marks)

- Sometimes spoilage micro-organisms can develop
- Acidity of the product is increased. **(1 Mark for each point.)**

13. Explain cooking methods in which air as the medium of cooking. [U] / [CO2]

- Grilling or broiling-**2 marks**
- Pan broiling-**2 marks**
- Baking-**1 mark**

14. Discuss about cleaning as a preliminary preparation method. [U] / [CO2]

- It includes discard the spoilt, withered or discolored parts of foods. -**1 mark**
- Washing- makes food dirt free, helps to remove husk and mud from rice, pesticide residues from fruits. -**1 mark**
- **Advantages of cleaning- 1 mark each for any 2 points**
  - Remove sprays, chemicals and dirt
  - Washing in warm water helps to remove worms
  - Helps to remove blood, dirt and unwanted impurities from flesh foods.
  - helps to remove gills from fish and hard shells from prawns and crabs
- **Disadvantage of cleaning-1 mark**
  - Washing causes loss of water-soluble nutrients.

15. Summarise the components and the process of standardization

Components of Recipe Standardization (2.5 marks)

Recipe Title
The title should be clear, descriptive, and appealing, highlighting key ingredients or a unique feature of the dish. For example, instead of "Dal," use "Authentic Punjabi Dal Tadka."

Recipe Category
Classifying recipes (main course, side dish, dessert, snack) helps organize menus and plan meals efficiently. For example, "Vegetable Biryani" is a main course, while "Cucumber Raita" is a side dish.

Ingredients List
List ingredients in order of use with exact names, form, and preparation method for consistency. For example, "250 g paneer, crumbled" or "1 tbsp cumin seeds, roasted."

Weight/Volume of Ingredients
Use precise standard units like grams or milliliters to ensure accuracy and consistency. For example, "500 g wheat flour" instead of "one bowl of flour."

Preparation Instructions
Provide clear, step-by-step instructions including techniques and pre-preparation steps. For example, "Marinate chicken with yogurt and spices for 4 hours before grilling."

Cooking Temperature and Time
Mention exact temperature and duration to ensure proper cooking and safety. For example, "Bake at 180°C for 25 minutes until golden brown."

Serving Size
Define portion size clearly for consistency and nutrition control. For example, "One serving = 200 g dal with 150 g rice."

Recipe Yield
State the total output in servings or quantity to help scaling and planning. For example, "Yields 10 servings of 250 g each."

Equipment and Utensils
List required tools and their size/type to ensure proper preparation. For example, "Use a 12-inch skillet and a 4 L mixing bowl."

PROCESS OF STANDARDISATION (2.5 marks)

I. Recipe Verification

Recipe verification ensures accuracy in ingredients, steps, and yield before regular use, helping maintain consistency and quality. For example, checking if "Vegetable Pulao" gives the correct number of servings.

1. Review the Recipe
Ensure the recipe has a clear title, proper category, accurate ingredients, and detailed instructions with correct units. For example, "250 g chopped carrots" instead of "some carrots."

2. Prepare the Recipe
Cook the recipe as written to check taste, texture, and ease of preparation. For example, note if cooking time needs adjustment for proper softness of rice.

3. Verify Recipe Yield
Measure the final output and calculate portion size to ensure consistency. For example, 5000 g total ÷ 25 servings = 200 g per portion.

4. Record Changes
Document any changes in ingredients, method, or equipment for future consistency. For example, reducing salt from 10 g to 8 g after testing.

II. Product Evaluation

Product evaluation checks whether a recipe is acceptable or needs improvement based on quality and feedback.

1. Informal Evaluation
The recipe is tested casually to check if it meets basic standards. For example, tasting a curry to see if flavor is balanced.

2. Formal Evaluation
A structured test is conducted with a group using evaluation forms to assess quality. For example, 25 people rating taste, texture, and appearance.

3. Collecting and Analyzing Feedback
Feedback is analyzed to decide whether to accept, modify, or reject the recipe. For example, low scores in taste may require spice adjustment.

III. Quantity Adjustment

This step adjusts recipe quantities to meet the required number of servings while maintaining consistency.

1. Factor Method
Multiply each ingredient by a conversion factor to scale the recipe. The Factor Method is a widely used approach for adjusting recipe quantities by multiplying each ingredient by a conversion factor. This method ensures that the ingredient proportions remain the same while scaling the recipe up or down to meet the required yield. For example, increasing 100 servings to 250 servings → factor = 2.5.

2. Direct Table Reading Method
Use pre-calculated tables to adjust ingredient quantities quickly without calculations. For example, combining 100-serving and 25-serving values to get 125 servings. Identify the portion size closest to the desired yield. If the desired yield is not directly available in the table, combine values from multiple entries (e.g., for 125 servings, use values from the 100-serving and 25-serving columns).

3. Percentage Method
Convert ingredients into percentages of total weight to adjust quantities accurately. The Percentage Method involves converting all ingredient measurements into weights and calculating the percentage of each ingredient relative to the total recipe weight. For example, sugar = 25% of 500 g → for 750 g, sugar = 187.5 g.

16. Explain the components of standardization.

Components of Recipe Standardization

Recipe Title (1 mark)
The title should be clear, descriptive, and appealing, highlighting key ingredients or a unique feature of the dish. For example, instead of "Dal," use "Authentic Punjabi Dal Tadka."

Recipe Category (1/2 mark)
Classifying recipes (main course, side dish, dessert, snack) helps organize menus and plan meals efficiently. For example, "Vegetable Biryani" is a main course, while "Cucumber Raita" is a side dish.

Ingredients List (1/2 mark)
List ingredients in order of use with exact names, form, and preparation method for consistency. For example, "250 g paneer, crumbled" or "1 tbsp cumin seeds, roasted."

Weight/Volume of Ingredients (1/2 mark)
Use precise standard units like grams or milliliters to ensure accuracy and consistency. For example, "500 g wheat flour" instead of "one bowl of flour."

Preparation Instructions (1/2 mark)
Provide clear, step-by-step instructions including techniques and pre-preparation steps. For example, "Marinate chicken with yogurt and spices for 4 hours before grilling."

Cooking Temperature and Time (1/2 mark)
Mention exact temperature and duration to ensure proper cooking and safety. For example, "Bake at 180°C for 25 minutes until golden brown."

Serving Size (1/2 mark)
Define portion size clearly for consistency and nutrition control. For example, "One serving = 200 g dal with 150 g rice."

Recipe Yield (1/2 mark)
State the total output in servings or quantity to help scaling and planning. For example, "Yields 10 servings of 250 g each."

Equipment and Utensils (1/2 mark)
List required tools and their size/type to ensure proper preparation. For example, "Use a 12-inch skillet and a 4 L mixing bowl."

Part D
Essay Questions
Answer any **2** questions
Each question carries **10** marks

17. Explain different types of solar cookers. Write about the advantages and disadvantages of solar cooking.

Types – 4 marks

- Box Cookers (Solar Box Ovens)
- Parabolic Cookers
- Panel Cookers
- Solar Ovens (Cooker Ovens)

Advantages of Solar Cooking (Any 3 points.1 mark each)

Environmentally Friendly:

- No harmful emissions, reducing pollution.
- Does not rely on non-renewable resources like fossil fuels.

Sustainable:

- Solar energy is free and inexhaustible, making solar cooking a long-term, renewable energy source.

Cost-Effective:

- After the initial setup, solar cookers incur no additional costs as they do not require fuel or electricity

Health Benefits:

- Cooks food at lower temperatures, preserving more nutrients compared to traditional cooking methods.
- No harmful fumes or smoke, which makes it safer for health.

No Fuel Dependency:

- Does not rely on gas, electricity, or firewood, which is particularly beneficial in areas with limited access to these resources.

Disadvantages of Solar Cooking (Any 3 points.1 mark each)

Weather Dependency:

- Solar cooking is only effective on sunny days. It doesn't work well on cloudy, rainy, or overcast days.
- **Slow Cooking Time:**
- Solar cookers generally cook food more slowly than traditional cooking methods, which may not be ideal when time is limited.

Limited Capacity:

- Many solar cookers can only handle small quantities of food at a time, making them less suitable for large meals.

Requires Space:

- Some solar cookers, especially larger models, require more space for setup and operation.

  Initial Cost:

- The cost of purchasing or building a solar cooker may be a barrier in some areas, although it's often offset by the long-term savings on fuel.

18. Describe on microwave cooking. [U] / [CO2

- Microwave cooking is a method of cooking food using electromagnetic waves from a power source magnetron.
- It works on dielectric heating, where polar molecules (mainly water) vibrate rapidly when exposed to microwaves, producing heat.
- Heat is produced inside the food, unlike conventional cooking where heat moves from outside to inside.
- The efficiency of microwave cookery depends upon the constituents of food being cooked.
- Water in food converts microwaves to heat energy very efficiently followed by starch, cellulose and protein. Least effective is fatty tissues.
- Only microwave-safe materials like glass, ceramic, and certain plastics can be used; metals reflect microwaves.**(1 mark for each point)**
- Any 2 advantages of microwave cooking **(1 mark for each point)**
- Any 2disadvantages of microwave cooking **(1 mark for each point)**

19. Explain the types of chemical changes and denaturation of protein.

Chemical Changes in Food (5 marks)

Chemical changes alter the composition of food, forming new substances with different properties and are usually irreversible. For example, cooking rice changes its texture and structure permanently.

Types of Chemical Changes

1. Cooking and Baking: Heat causes reactions that change food's texture, flavor, and appearance. For example, baking bread makes it soft and fluffy, and heating sugar leads to caramelization.

2. Fermentation: Microorganisms break down food components to produce new flavors and textures. For example, milk turns into curd due to lactic acid bacteria.

3. Oxidation: Oxygen reacts with food, causing changes in color, taste, or quality. For example, cut apples turn brown due to enzymatic oxidation.

4. Enzymatic Reactions: Natural enzymes speed up chemical changes in food. For example, fruits become sweeter and softer during ripening.

Denaturation of Proteins (5 marks)

1. Definition: Denaturation is the loss of a protein's natural structure due to factors like heat or acids, without breaking peptide bonds. For example, egg white turns solid when cooked.

Causes of Protein Denaturation

1. Heat : High temperature unfolds proteins by breaking bonds. For example, cooking eggs turns liquid egg white into a solid.

2. Acids and Bases: pH changes disrupt protein stability. For example, adding lemon juice to milk causes curdling.

3. Mechanical Action: Beating or whisking changes protein structure. For example, whipping egg whites forms foam.

4. Salts and Heavy Metals: High salt or metals interfere with protein interactions. For example, salt curing changes meat texture.

5. Alcohol: Alcohol breaks protein bonds, causing denaturation. For example, sanitizers kill bacteria by denaturing proteins.

20. Describe standardisation, objectives and purpose.

Standardized Recipe (2 marks)

A standardized recipe is one that has been tested multiple times and produces the same quality and yield under the same conditions. For example, a recipe that always gives 10 servings of identical-tasting "Vegetable Pulao." According to USDA, it is a recipe that consistently gives the same results in quality and quantity after repeated trials. For example, a soup recipe that always yields 2 liters with the same taste. Recipe standardization is the process of fixing ingredients, measurements, methods, and portion sizes to ensure consistency. For example, using exactly 200 g rice and 500 ml water each time.

OBJECTIVES (3 marks)

1. To determine the quantity and quality of the ingredient be used.
2. To obtain the yield obtainable from a recipe.
3. To determine the cost per portion.
4. To determine the nutritional value.
5. To facilitate portion control.
6. It helps in costing of dishes, pricing menus for the banquet.
7. It helps in uniform quality and taste.
8. Require less supervision.
9. Less training is required for a newly appointed employee.
10. Establishes food cost control

Purpose of Recipe Standardization (5 marks)

1. Ensuring Uniformity in Food Preparation
Standardized recipes ensure consistent taste, texture, and appearance every time. For example, a cake with the same softness and sweetness in every batch.

2. Improving Operational Efficiency
Clear instructions and exact measurements make cooking faster and reduce errors. For example, staff can prepare dishes quickly during peak hours.

3. Aiding Staff Training
They help new employees learn correct methods easily without relying on experience. For example, a trainee following a step-by-step biryani recipe.

4. Managing Resources Effectively
Fixed recipes prevent wastage and ensure proper use of ingredients. For example, using only required oil instead of excess.

5. Meeting Nutritional Standards
They ensure meals meet required nutritional values in institutions. For example, hospital diets providing controlled calories and nutrients.